

# **Course. Introduction to Machine Learning**

## **Lecture 2. Introduction to unsupervised learning and Cluster Analysis (Part I)**

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# Introduction to Machine Learning

## Unsupervised Learning

## Supervised Learning

## Decision Learning Theory

### Cluster Analysis

Factor Analysis

Visualization

Non Linear Decision

Linear Decision

Basic concepts of Decision Learning Theory

### K-Means, Fuzzy C-means EM

PCA, ICA

Self Organized Maps (SOM), Multi-Dimensional Scaling

Lazy Learning (K-NN, IBL, CBR)

Overfitting, model selection and feature selection

Kernel Learning

Ensemble Learning (NN, Trees, Adaboost)

Perceptron, SVM

Bias/Variance, VC dimension, Practical advice of how to use learning algorithms

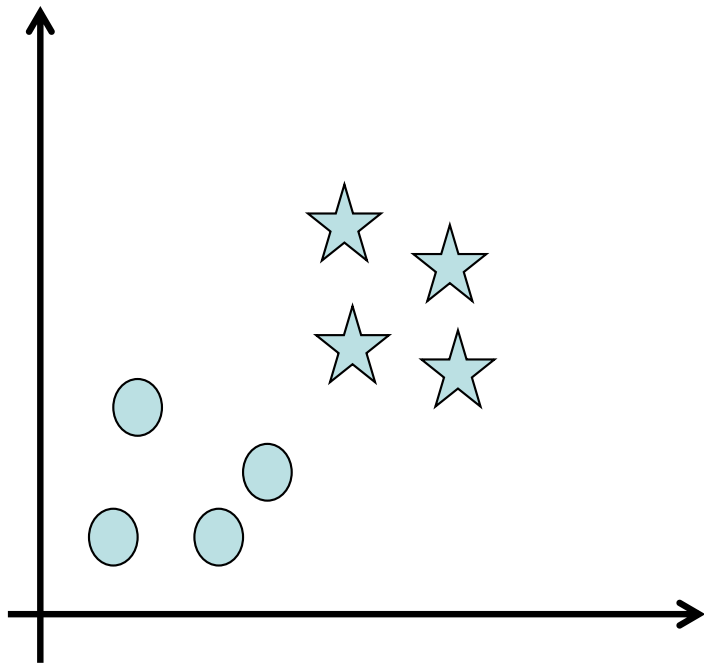
## 1. Introduction to unsupervised learning

1. Introduction to unsupervised learning
2. Examples
3. Definition of unsupervised learning
4. Unsupervised learning approaches

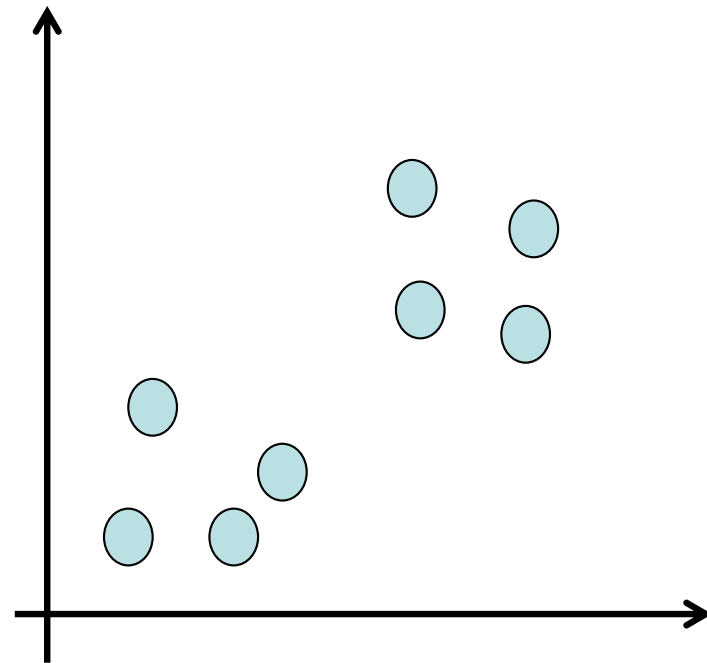
## 2. Cluster analysis

1. Defining clustering analysis
2. Areas that apply clustering
3. Classification of clustering algorithms
4. K-Means algorithm, bisecting K-Means, Fuzzy C-Means
5. EM (expectation maximization algorithm)
6. Clustering validation
7. Applications

*Supervised*



*Unsupervised*



- **Main Goals:**
  - **Summarization:** To obtain representations that describe an unlabeled dataset
  - **Understanding:** To discover the key concepts inside the data
- These tasks are difficult because the discovery process is biased by context
  - Different answers can be valid depending of the discovery goal or the domain
  - There are few criterion to validate the results
- **Representation of the clusters:**
  - Unstructured (partitions)
  - Relational (hierarchies)

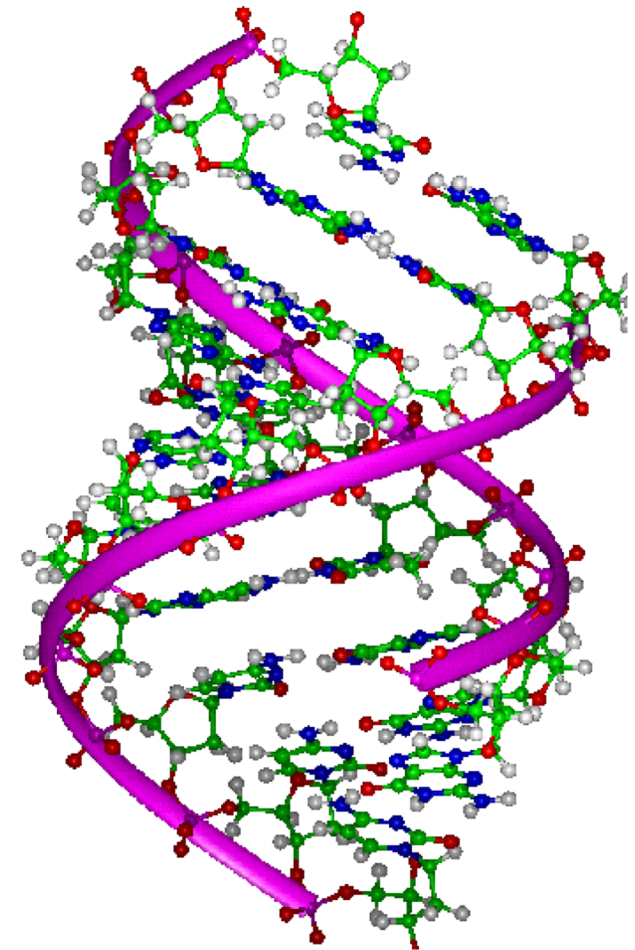
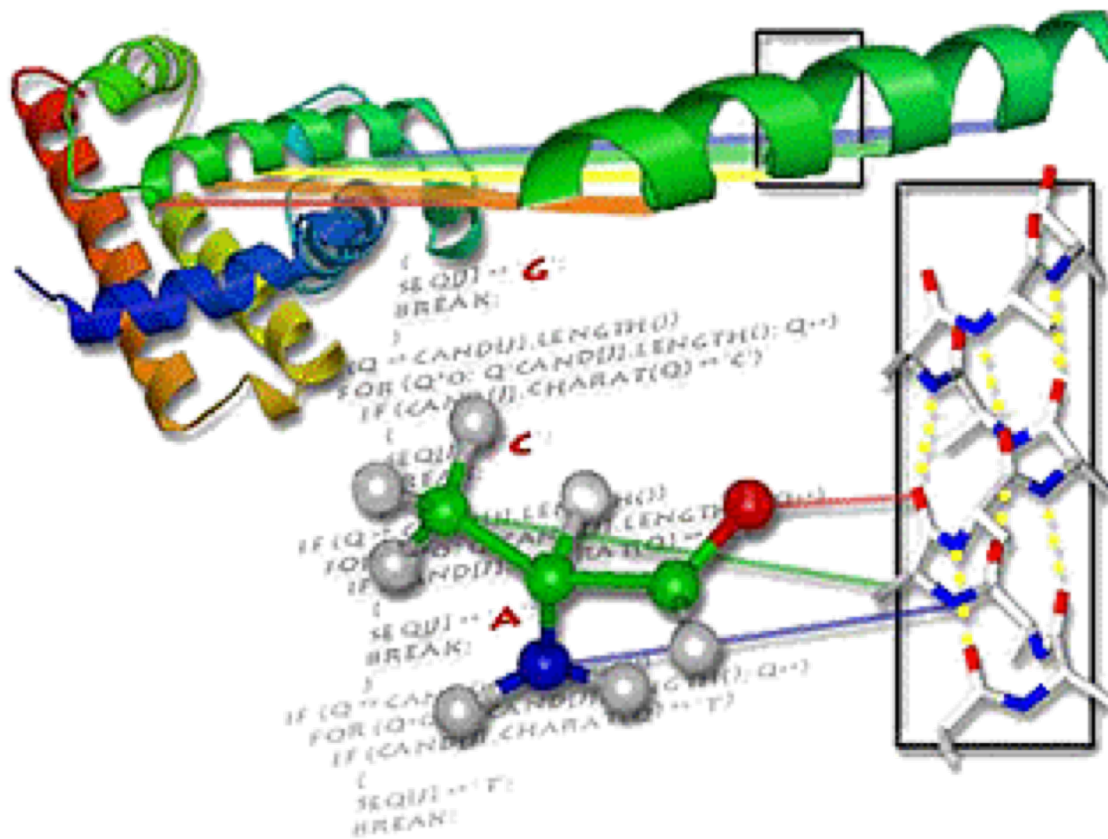
- Bioinformatics
- Medicine
- Market research
- Social network analysis
- NLP: document clustering, text mining, concept extraction
- Image segmentation
- Educational data mining
- Climatology
- ...



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# Bioinformatics



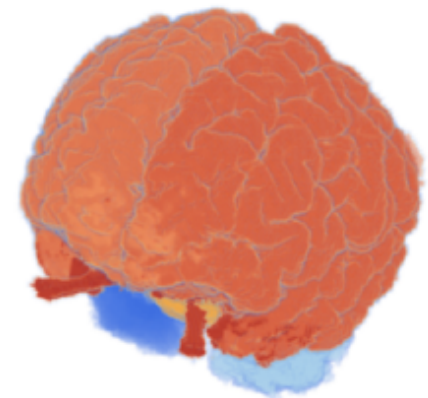
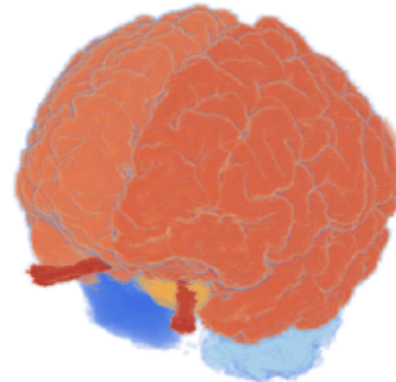
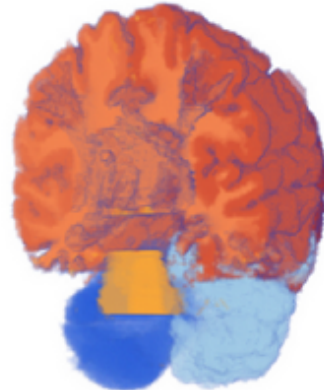
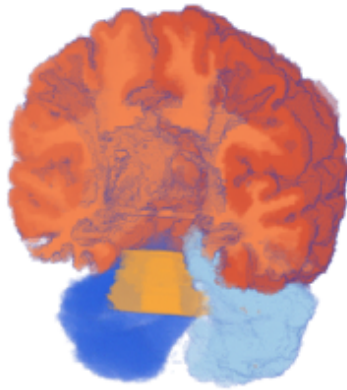
*Illustration by Brian Haas, Phillips lab*



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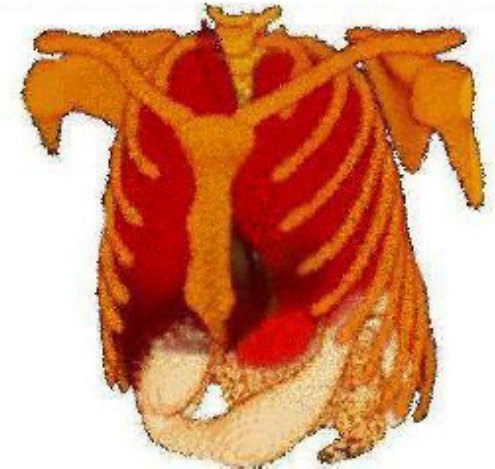
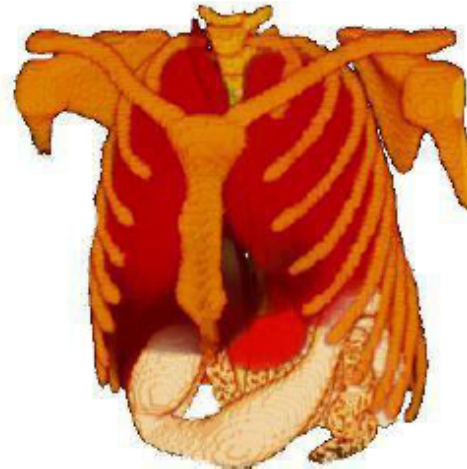
# Medicine



(a)



(b)



(c)



# Market research



# Social network analysis



- **Clustering**
  - It is the task of grouping a set of objects in such a way that objects in the same group (called a **cluster**) are more similar (in some sense or another) to each other than to those in other groups (clusters)
- **Factor Analysis**
  - Statistical methods used to describe variability among observed, correlated variables in terms of a potentially lower number of unobserved variables called factors.
- **Visualization**
  - Study of (interactive) visual representations of abstract data to reinforce human cognition.

- **Clustering:** method by which large sets of data is grouped into clusters of smaller sets of similar data
  - **Based on connectivity:** Hierarchical clustering
  - **Based on centroids:** K-means
  - **Distribution-based models:** Mixture models, Expectation-Maximization
  - Density models: DBScan, Optics
  - Subspace models: Biclustering
  - Group models
  - Graph-based models

- **Factor analysis:** blind signal separation using *feature extraction* techniques for *dimensionality reduction*
  - Principal components analysis (**PCA**)
  - Independent component analysis (**ICA**)
  - Non-negative matrix factorization
  - Singular value decomposition (SVD)
  - ...

- **Visualization:** a set of techniques often used in **information visualization** for exploring similarities and dissimilarities in data
  - **Neural network models:**
    - **Self-organized maps (SOM)**
    - Adaptive resonance theory (ART)
  - **Multi-dimensional scaling (MDS)**
    - **Classical multidimensional scaling**
    - Metric multidimensional scaling
    - Non metric multidimensional scaling
    - Generalized multidimensional scaling

## 1. Introduction to unsupervised learning

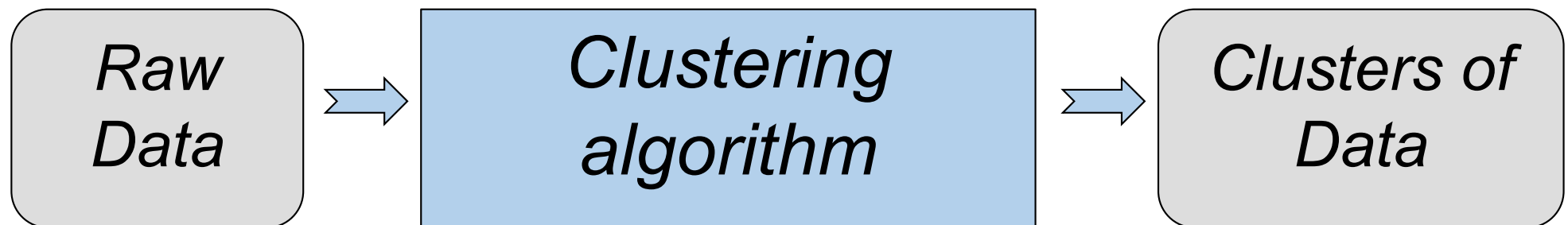
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## 2. Cluster analysis

1. Defining clustering analysis
2. Areas that apply clustering
3. Classification of clustering algorithms
4. K-Means
5. EM
6. Other algorithms
7. Applications

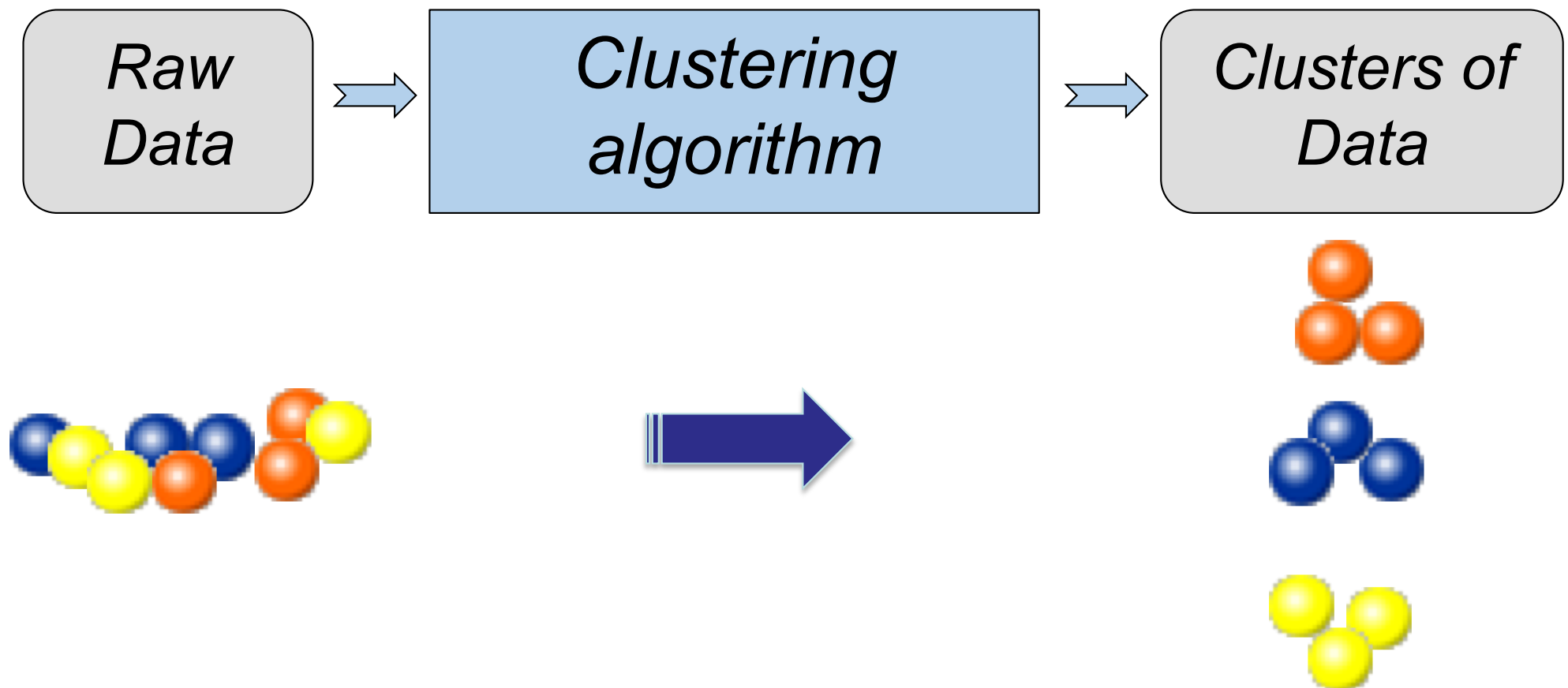
# Defining clustering analysis

- **Cluster analysis** or **clustering** is the task of assigning a set of unlabelled objects into groups (**clusters**) so that the objects in the same cluster are very similar (in some sense or another) to each other than those of the other clusters.
- Cluster analysis discover new categories in an **unsupervised** manner



# Defining clustering analysis

- A clustering algorithm attempts to find natural groups of components (or data) based on some similarity.



Clustering is ...

- Main task of explorative **data mining**
- Common technique for **statistical data analysis** used in many fields:
  - Machine learning
  - Pattern recognition
  - Information retrieval
  - Bioinformatics
  - Natural language processing
  - Recommender systems
  - Data mining
  - ...

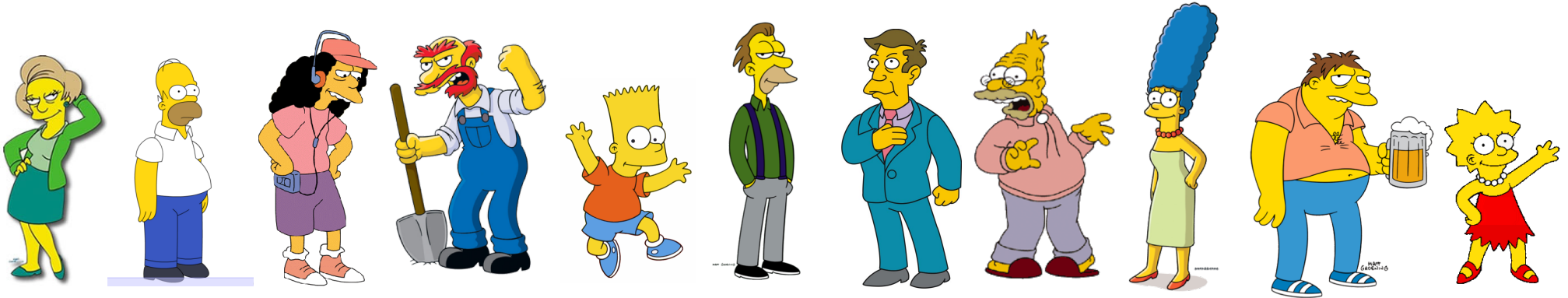
# Clustering Example



*How many clusters?*

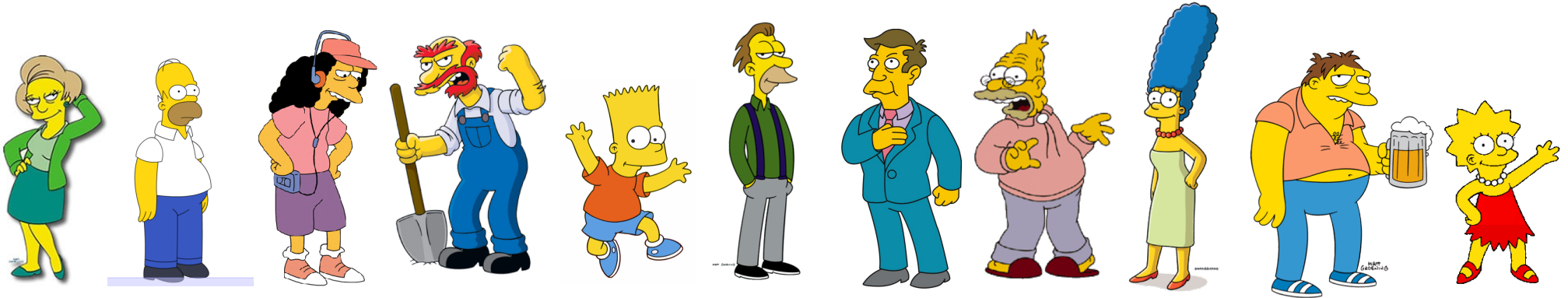
- What is a natural grouping among these objects?
  - Definition of “groupness”
- What makes object “related”?
  - Definition of “similarity/distance”
- Representation of objects
  - Vector space? Normalization?
- How many clusters
  - Fixed a priori?
  - Completely data driven?
- Clustering algorithms
  - Hierarchical algorithms
  - Partitional algorithms
- Formal foundation and convergence

# What is a natural grouping?



***Decide how to group them !!!***

# What is a natural grouping?



*Decide how to group them, in a different way !!!*



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# What is similarity?

- $D(A, B) = D(B, A)$  Symmetry
- $D(A, A) = 0$  Constancy of Self-Similarity
- $D(A, B) = 0 \mid \text{if } A=B$  Positivity Separation
- $D(A, B) \leq D(A, C) + D(B, C)$  Triangular Inequality

- Suppose two object  $x$  and  $y$  both have  $p$  features

$$x = (x_1, x_2, \dots, x_p)$$

$$y = (y_1, y_2, \dots, y_p)$$

- The Minkowski metric is defined by

$$d(x, y) = \sqrt[r]{\sum_{i=1}^p |x_i - y_i|^r}$$



# Distance measures:

## Pearson correlation

- Pearson correlation coefficient:

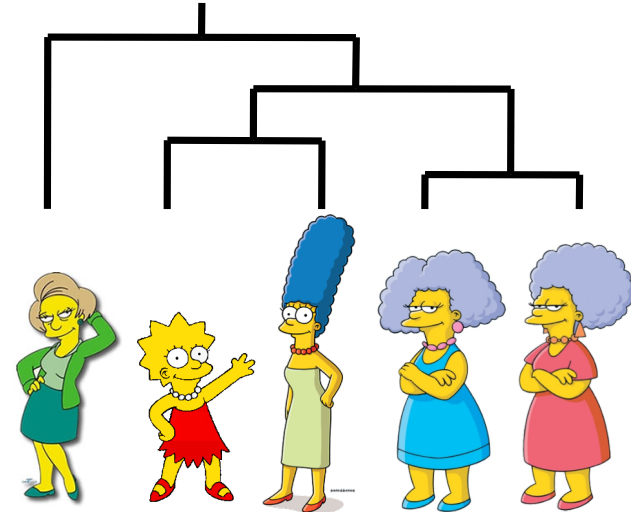
$$s(x, y) = \frac{\sum_{i=1}^p (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^p (x_i - \bar{x})^2 \times \sum_{i=1}^p (y_i - \bar{y})^2}}$$

$$\text{where } \bar{x} = \frac{1}{p} \sum_{i=1}^p x_i \text{ and } \bar{y} = \frac{1}{p} \sum_{i=1}^p y_i.$$

$$|s(x, y)| \leq 1$$

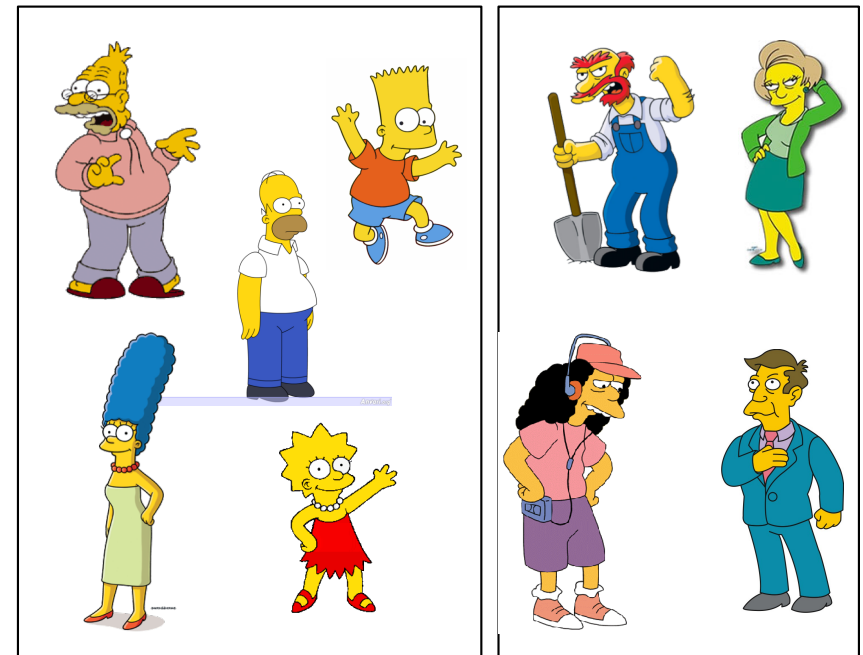
- **Hierarchical algorithms**

- Examples are organized as a binary tree
- No explicit division in groups
  - Bottom-up
  - Top-down



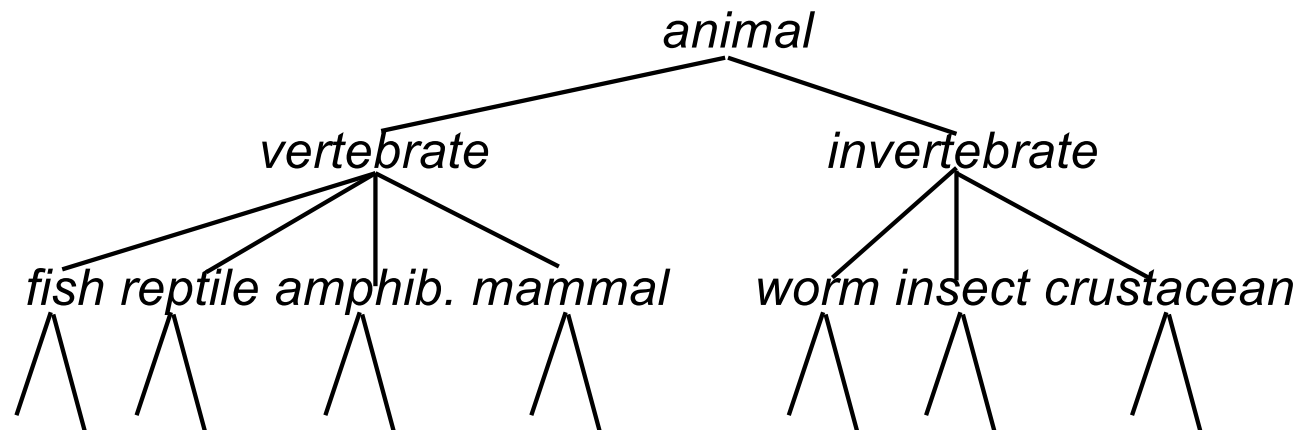
- **Partitional algorithms**

- Usually start with a random (partial) partitioning
- Refine it iteratively:
  - K-means clustering
  - Mixture-model based clustering



# Hierarchical Clustering

- Build a tree-based hierarchical taxonomy (*dendrogram*) from a set of unlabeled examples



- Recursive application of a standard clustering algorithm can produce a hierarchical clustering

- *Agglomerative (bottom-up)*
  - Methods start with each example in its own cluster
  - Iteratively combine them to form larger and larger clusters
- *Divisive (partitional, top-down)*
  - Methods start with all the examples in a single cluster
  - Consider all the possible way to divide the cluster into two. Choose the best division
  - Recursively operate on both sides

## Basic HAC algorithm:

1. Compute the similarity matrix between the input data points
2. Start with all instances in their own cluster
3. **Repeat**
4.       Among the current clusters, determine the two clusters,  $c_i$  and  $c_j$ , that are most similar
5.       Merge them and replace  $c_i$  and  $c_j$  with a single cluster  $c_i \cup c_j$
6.       Update the similarity matrix
7. **until** there is only one single cluster

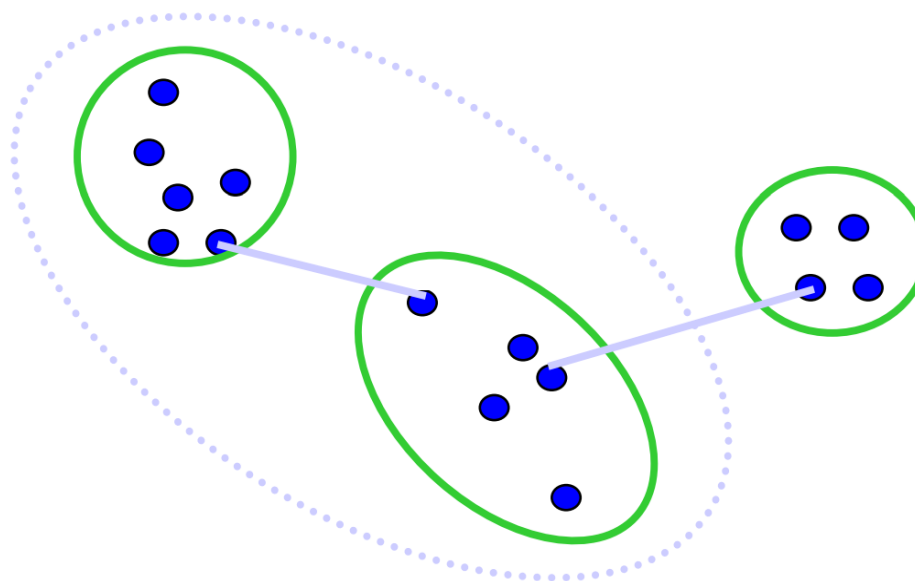
- **Key operation is the computation of the similarity between two clusters**
  - Different definitions of the similarity between clusters lead to different algorithms

- Assume a similarity function that determines the similarity of two instances: ***sim(x,y)***
  - For example, Cosine similarity of document vectors
- How to compute similarity of two clusters each possibly containing multiple instances?
  - **Single Link**: Similarity of two most similar members
  - **Complete Link**: Similarity of two least similar members
  - **Group Average**: Average similarity between members
  - **Centroid**: clusters whose centroids are the most cosine similar

- Use maximum similarity of pairs:

$$\text{sim}(c_i, c_j) = \max_{x \in c_i, y \in c_j} \text{sim}(x, y)$$

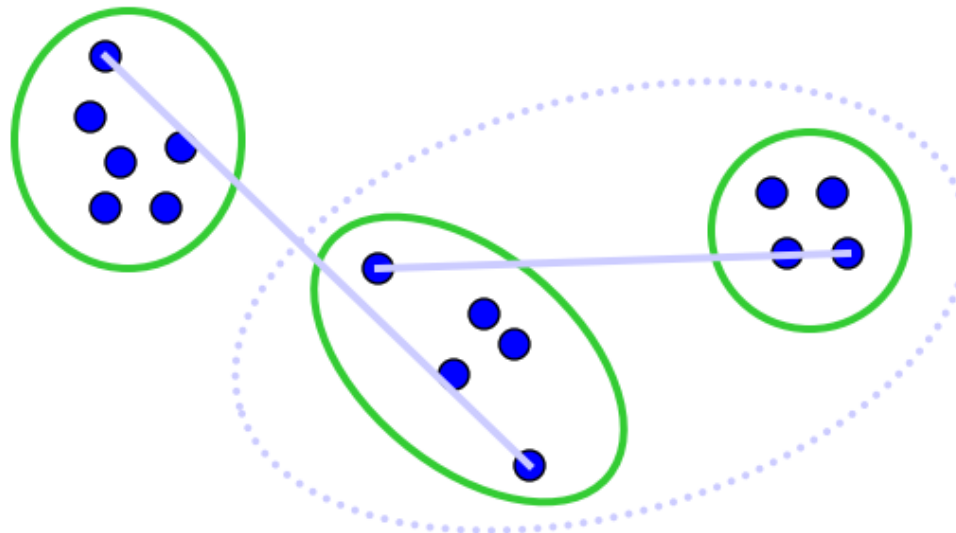
- Can result in “straggly” (long and thin) clusters due to *chaining effect*.
  - Appropriate in some domains, such as clustering islands



- Use minimum similarity of pairs:

$$\text{sim}(c_i, c_j) = \min_{x \in c_i, y \in c_j} \text{sim}(x, y)$$

- Makes more “tight,” spherical clusters that are typically preferable



- In the first iteration, all HAC methods need to compute similarity of all pairs of  $n$  individual instances which is  $O(n^2)$ .
- In each of the subsequent  $n-2$  merging iterations, it must compute the distance between the most recently created cluster and all other existing clusters.
- In order to maintain an overall  $O(n^2)$  performance, computing similarity to each other cluster must be done in constant time.
- Else  $O(n^2 \log n)$  or  $O(n^3)$  if done naively

- After merging  $c_i$  and  $c_j$ , the similarity of the resulting cluster to any other cluster,  $c_k$ , can be computed by:

- **Single-Link:**

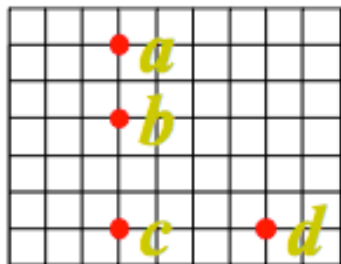
$$\text{sim}((c_i \cup c_j), c_k) = \max(\text{sim}(c_i, c_k), \text{sim}(c_j, c_k))$$

- **Complete-Link:**

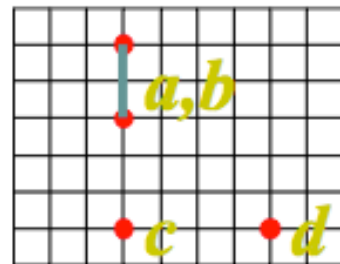
$$\text{sim}((c_i \cup c_j), c_k) = \min(\text{sim}(c_i, c_k), \text{sim}(c_j, c_k))$$

# Single-Link Example

## Euclidean Distance

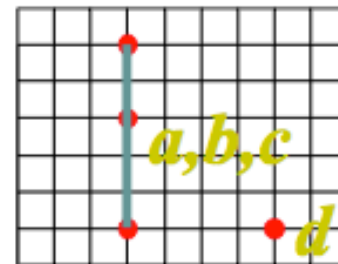


|          | <i>b</i> | <i>c</i> | <i>d</i> |
|----------|----------|----------|----------|
| <i>a</i> | 2        | 5        | 6        |
| <i>b</i> |          | 3        | 5        |
| <i>c</i> |          |          | 4        |



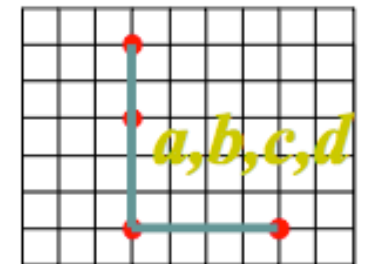
(1)

|          | <i>b</i> | <i>c</i> | <i>d</i> |
|----------|----------|----------|----------|
| <i>a</i> | 2        | 5        | 6        |
| <i>b</i> |          | 3        | 5        |
| <i>c</i> |          |          | 4        |



(2)

|             | <i>c</i> | <i>d</i> |
|-------------|----------|----------|
| <i>a, b</i> | 3        | 5        |
| <i>c</i>    |          | 4        |



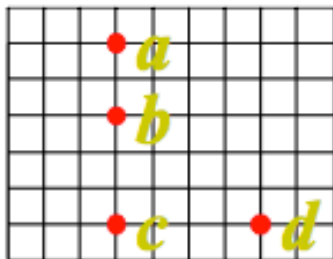
(3)

|                | <i>d</i> |
|----------------|----------|
| <i>a, b, c</i> | 4        |

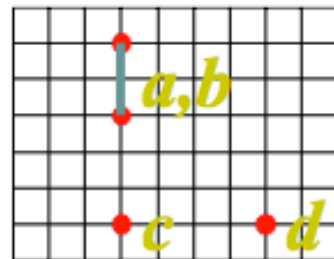
## Distance Matrix

# Complete-Link Example

## Euclidean Distance

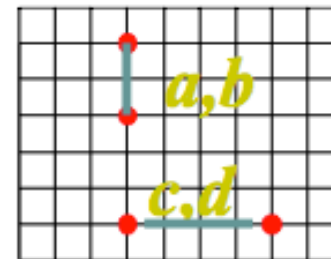


|          | <i>b</i> | <i>c</i> | <i>d</i> |
|----------|----------|----------|----------|
| <i>a</i> | 2        | 5        | 6        |
| <i>b</i> |          | 3        | 5        |
| <i>c</i> |          |          | 4        |



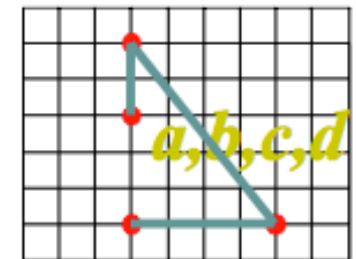
(1)

|          | <i>b</i> | <i>c</i> | <i>d</i> |
|----------|----------|----------|----------|
| <i>a</i> | 2        | 5        | 6        |
| <i>b</i> |          | 3        | 5        |
| <i>c</i> |          |          | 4        |



(2)

|             | <i>c</i> | <i>d</i> |
|-------------|----------|----------|
| <i>a, b</i> | 5        | 6        |
| <i>c</i>    |          | 4        |



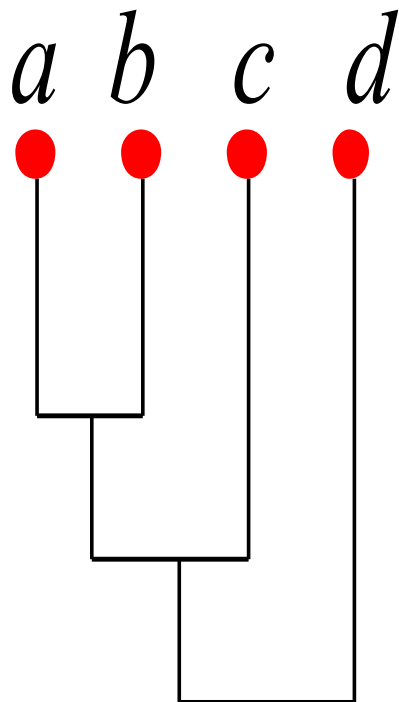
(3)

|             | <i>c, d</i> |
|-------------|-------------|
| <i>a, b</i> | 6           |

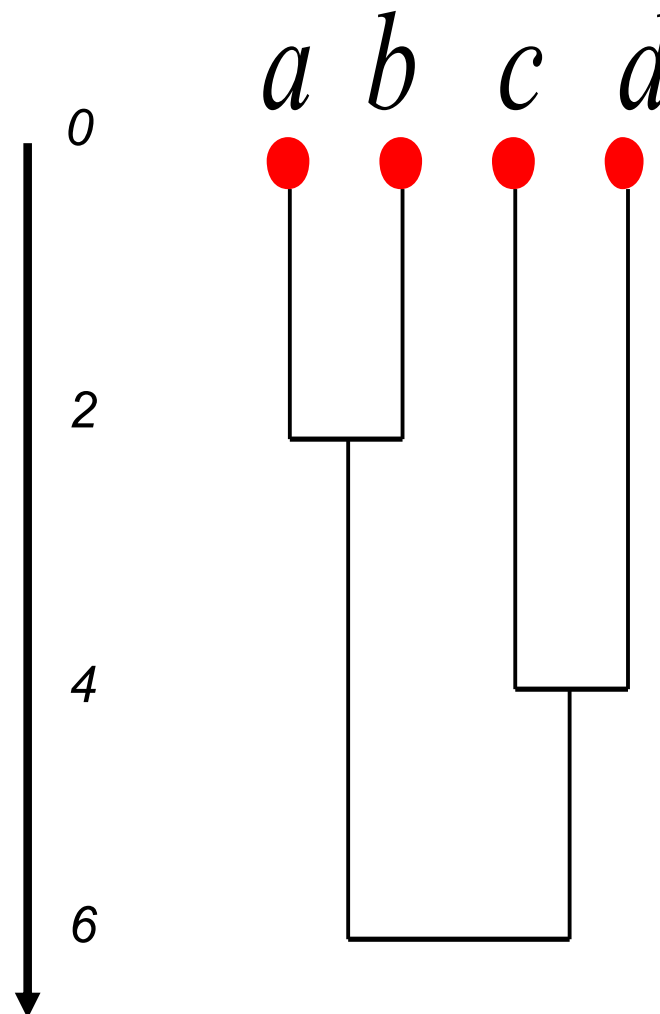
## Distance Matrix

# Dendrograms Comparison

*Single-Link*



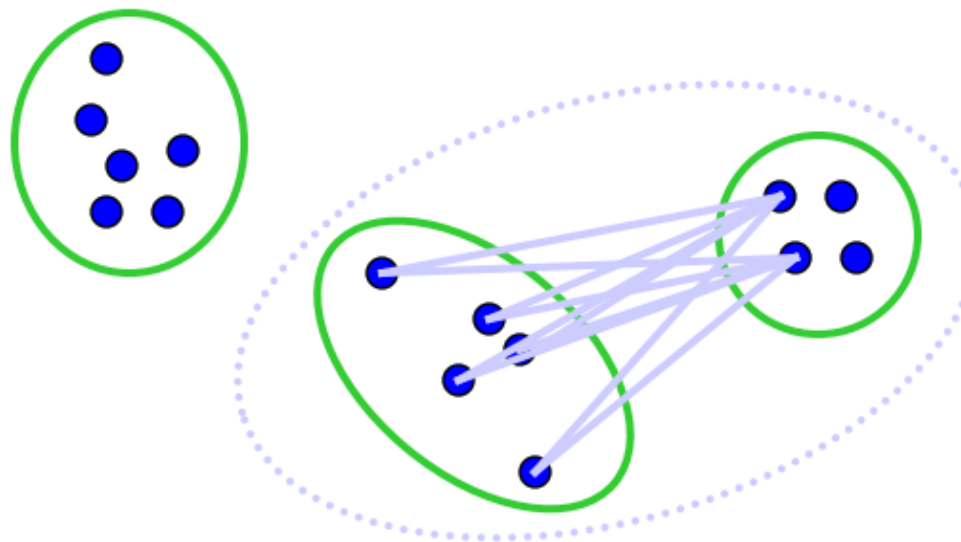
*Complete-Link*



- Use average similarity across all pairs within the merged cluster to measure the similarity of two clusters.

$$sim(c_i, c_j) = \frac{1}{|c_i \cup c_j|(|c_i \cup c_j| - 1)} \sum_{\vec{x} \in (c_i \cup c_j)} \sum_{\vec{y} \in (c_i \cup c_j): \vec{y} \neq \vec{x}} sim(\vec{x}, \vec{y})$$

- Compromise between Single and Complete link.



- Assume cosine similarity and normalized vectors with unit length.

- Always maintain sum of vectors in each cluster.

$$\vec{s}(c_j) = \sum_{\vec{x} \in c_j} \vec{x}$$

- Compute similarity of clusters in constant time:

$$\text{sim}(c_i, c_j) = \frac{(\vec{s}(c_i) + \vec{s}(c_j)) \bullet (\vec{s}(c_i) + \vec{s}(c_j)) - (|c_i| + |c_j|)}{(|c_i| + |c_j|)(|c_i| + |c_j| - 1)}$$